M.SC. STATISTICS · PROJECT REPORT

Credit Default Risk Scorecard

Predicting probability of default and building an interpretable, cost-aware credit scorecard

German Credit (Statlog) dataset · Weight-of-Evidence + Calibrated Logistic Regression · PDO scaling · Cost-sensitive cut-off

Prepared by	C. Shyam Sunder	**Tools**	Python · scikit-learn · pandas
Programme	M.Sc. Statistics	**Date**	June 2026
Dataset	UCI Statlog (German Credit Data) — 1,000 applicants, 30% defaults · CC BY 4.0		

1. Executive Summary

This project builds a credit scorecard that estimates how likely a loan applicant is to default, using the German Credit dataset of 1,000 applicants (30% of them defaults). It covers the full set of steps in a credit-risk project: cleaning and exploring the data, engineering a few new features, transforming features with Weight of Evidence (WoE) and screening them by Information Value (IV), training a logistic regression, choosing a decision cut-off from a cost rule, turning the model into a points-based scorecard, validating it, auditing it for fairness, and grouping applicants into risk bands.

On the 30% hold-out the calibrated logistic model gives **AUC 0.809, Gini 0.619 and KS 0.514**, with a train-to-test **PSI of 0.043** (a stable population). It performs on par with a HistGradientBoosting benchmark (AUC 0.786) — the small gap is within normal variation on a 300-row test set — so the simpler, easier-to-explain model is preferred at no real cost in performance. Choosing the cut-off from the dataset's 5:1 cost rule brings the expected cost down from 220 to 149 and raises the share of defaulters caught from 58% to 81%.

2. Data and Preparation

The raw *german.data* file is space-delimited with no header, so it is loaded with explicit column names and the target is recoded so that class 2 (bad credit) becomes 1 (default). The data mixes categorical attributes (checking-account status, credit history, loan purpose, savings, employment, property, housing) with numeric attributes (loan duration, credit amount, age, instalment rate). There are no missing values and no duplicate rows. The classes are imbalanced at roughly 70:30 (Figure 1), which drives three deliberate modelling choices downstream: accuracy is rejected as the headline metric in favour of AUC, KS and recall; the logistic model is fitted with balanced class weights; and the decision threshold is set from a cost rule rather than the default 0.5.

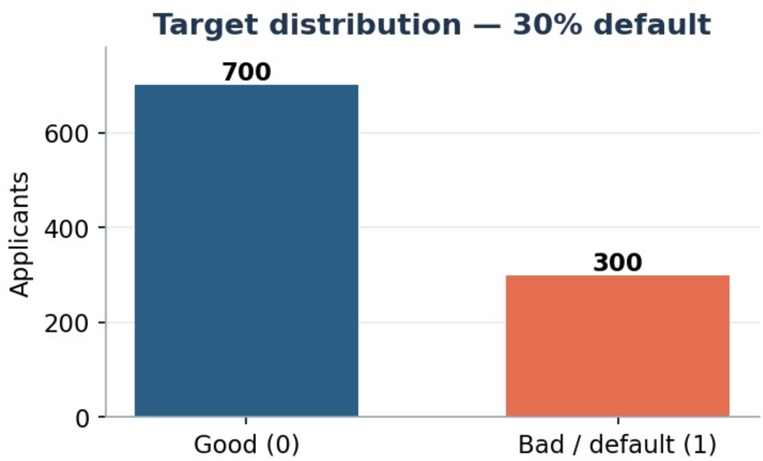

Figure 1 — Target distribution: 700 good versus 300 bad credits (30% default).

3. Exploratory Analysis: What Drives Default

Default rates were profiled across the strongest categorical drivers before modelling. The relationships align with established credit intuition, which is reassuring evidence that the eventual model captures genuine signal rather than noise. Checking-account status is the single most informative field: applicants with a negative balance default far more often than those with a healthy balance or no checking account at all, and a clean repayment history sharply lowers risk while past delays or critical credits raise it (Figure 2).

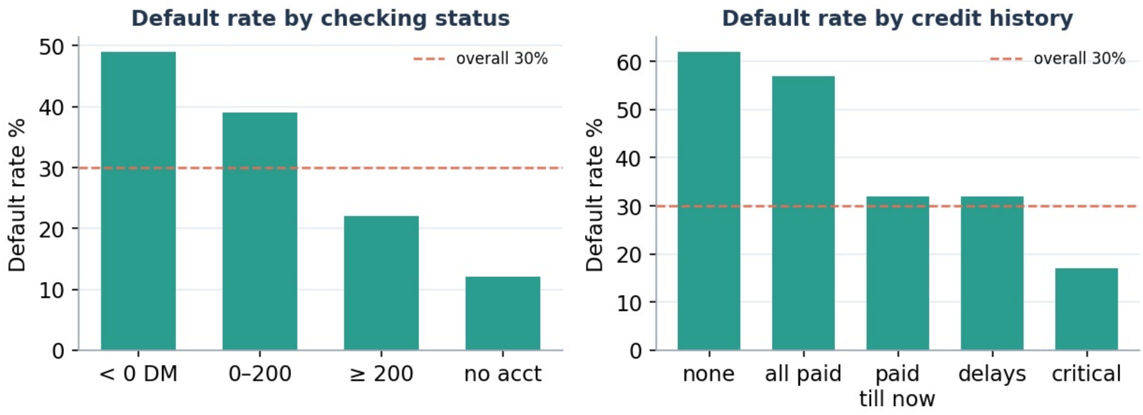

Figure 2 — Default rate by checking-account status (left) and credit history (right); dashed line is the 30% base rate.

Savings and loan duration tell an equally coherent story (Figure 3): higher savings reduce default risk, and longer loans are riskier because more can go wrong over a longer horizon. The savings effect is strong at the extremes and flatter in the middle — a non-linear shape that motivates WoE binning over a single linear term, since binning lets each range carry its own risk weight.

Figure 3 — Default rate by savings level (left) and loan duration (right).

4. Feature Engineering

Two ratio features were constructed from existing fields: *credit_per_month* (credit amount ÷ loan duration, a proxy for monthly repayment burden) and *amount_to_age* (credit amount relative to age, a proxy for loan size against life stage). Both rank among the stronger predictors by Information Value, confirming they add signal rather than noise. A correlation review showed every pairwise correlation among numeric features below the 0.85 pruning threshold, so each feature contributes distinct information — important for a logistic model, where near-duplicate inputs destabilise coefficients and undermine the interpretability the scorecard depends on.

5. Weight of Evidence and Information Value

Each feature is binned — a shallow decision tree for numeric fields, native levels for categoricals — and scored by Information Value with Laplace smoothing to avoid divide-by-zero. Critically, WoE bins are learned on the training fold only and then applied to the test fold, preventing target leakage. Features with IV below 0.02 are dropped and any WoE feature correlated above 0.85 with another is removed. The resulting ranking (Figure 4, Table 1) is led by checking-account status, followed by loan size, the engineered amount-to-age ratio and loan duration; no IV is implausibly high, which would have signalled leakage.

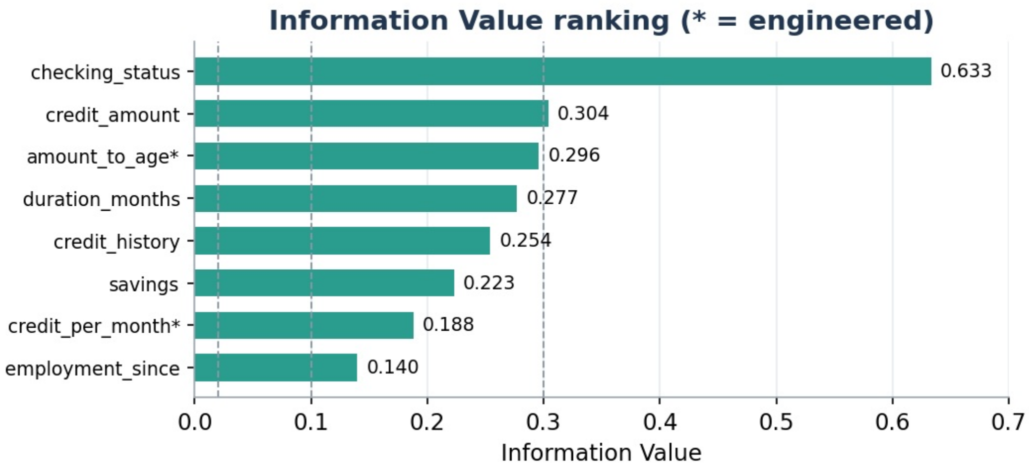

Figure 4 — Information Value ranking (top features). Dashed lines mark the 0.02 / 0.10 / 0.30 strength thresholds.

Feature	Information Value	Predictive strength
checking_status	0.633	Strong
credit_amount	0.304	Strong
amount_to_age *(engineered)*	0.296	Medium
duration_months	0.277	Medium
credit_history	0.254	Medium
savings	0.223	Medium
credit_per_month *(engineered)*	0.188	Medium
employment_since	0.140	Medium

The WoE profile for checking-account status (Figure 5) increases monotonically from the riskiest bin (negative balance) to the safest (no account / healthy balance). A clean, ordered shape like this is exactly what a scorecard needs: a worse status simply costs the applicant points, with no jumpy or out-of-order weights that would signal poor binning.

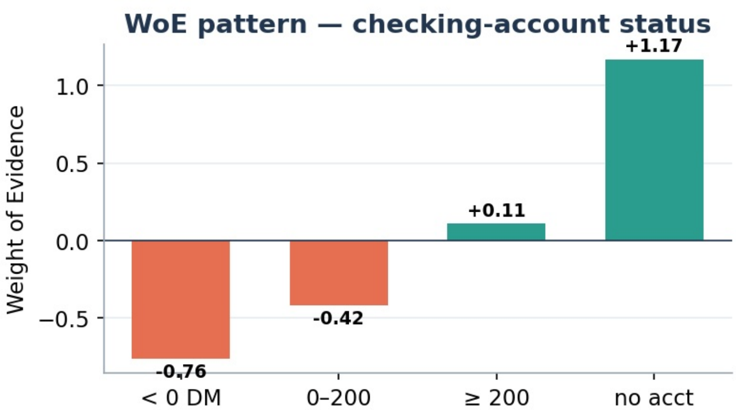

Figure 5 — Weight-of-Evidence pattern for checking-account status (negative values = higher risk).

6. Modelling and Calibration

The data is split 70/30 with stratification to keep the same good/bad mix in both parts. A logistic regression with balanced class weights is trained on the WoE features, then isotonic calibration is added so the predicted probabilities line up with the actual default rates. Calibration matters because a model can rank applicants well yet still be consistently too high or too low — fine for ranking, but a problem if the scores are ever used for pricing or expected-loss work. After calibration, a predicted 20% default risk corresponds to roughly 20% observed defaults.

7. Cost-Sensitive Decisioning

A 0.5 cut-off maximises raw accuracy but lets too many defaulters through. The German Credit dataset ships with an official cost matrix in which approving a bad applicant (a false negative) costs five times more than declining a good one (a false positive). The threshold is therefore chosen to minimise total expected cost. Three operating points are compared

(Table 2, Figure 6); the cost-optimal cut-off of 0.25 is deployed.

Configuration	Threshold	Recall	Precision	F1	Expected cost
Naive cut-off	0.50	0.578	0.634	0.605	220
Youden's J (KS point)	0.285	0.767	0.566	0.651	158
Cost-sensitive 5:1 (selected)	0.25	0.811	0.533	0.643	**149**

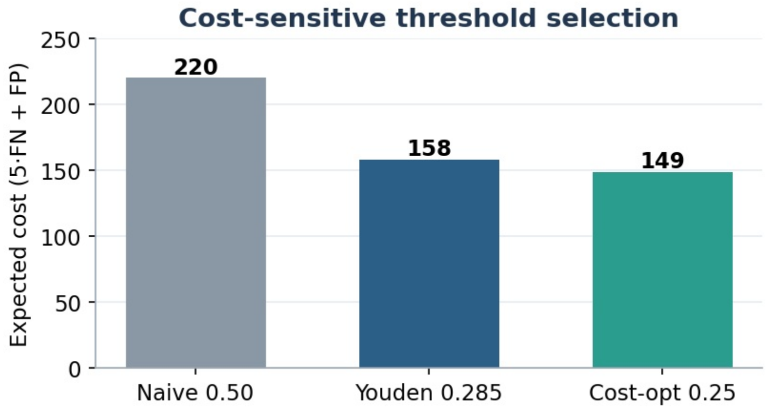

Figure 6 — Expected cost (5·FN + FP) by operating point; the cost-optimal cut-off sits well below 0.5.

Moving from 0.5 to 0.25 cuts expected cost by roughly a third (220 → 149) and raises the share of true defaulters caught from 58% to 81%. The trade-off is lower precision — more good applicants are routed to manual review — which is economically justified precisely because a missed default is five times costlier than an unnecessary review.

8. Validation and Benchmark

On the held-out test set the model is assessed with banking-standard metrics and compared against a HistGradientBoosting benchmark (Table 3). The confusion matrix at the deployed cut-off (Figure 7) shows the intended lean toward catching defaulters: of 90 true bad credits, 73 are caught.

Metric	Value	Reading
AUC	0.809	0.5 = random
Gini	0.619	2·AUC − 1
KS	0.514	> 0.30 = good separation
PSI (train→test)	0.043	< 0.10 = stable
Benchmark GBM AUC	0.786	see note

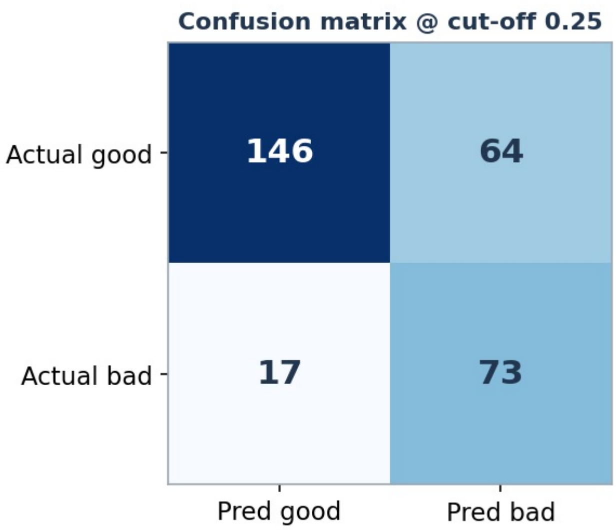

Figure 7 — Confusion matrix at cut-off 0.25.

On the benchmark comparison. The logistic model (AUC 0.809) is **on par with, not clearly better than**, the gradient-boosting model (AUC 0.786). On a 300-row test set a gap of about 0.02 AUC is within normal variation, so it should not be read as higher accuracy. The reason to prefer the logistic scorecard is that it matches a stronger non-linear model while staying simple and easy to explain — which is what a credit model is usually chosen for.

9. Credit Scorecard (PDO Scaling)

Calibrated probabilities are converted to points with Factor = PDO / ln 2 and an offset anchored to the population's average log-odds (PDO 20, base score 600, base odds 50:1). Resulting scores range from 469 to 938 with a mean of 593, where a higher score means lower risk. Anchoring the offset to the population average keeps every risk band populated — avoiding the empty-band pitfall common in naive scorecards. Table 4 shows representative weights from the full 82-row points table.

Feature	Bin	WoE	Risk direction
checking_status	Negative balance (A11)	−0.76	Higher risk
checking_status	No account (A14)	+1.17	Lower risk
credit_history	Critical / other credits (A34)	positive	Lower risk
duration_months	Long-term loans	negative	Higher risk

10. Risk Segmentation and Policy

Scores are bucketed into high / medium / low risk bands. The default rate falls monotonically across bands — about a six-fold drop from the high-risk to the low-risk band (Figure 8, Table 5) — which maps cleanly onto a three-way lending policy.

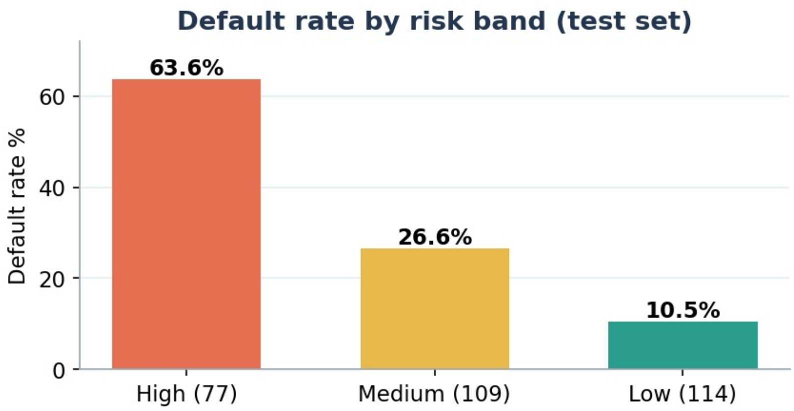

Figure 8 — Default rate by risk band on the test set; all bands populated and correctly ordered.

Risk band	Applicants	Defaults	Default rate	Recommended action
High	77	49	63.6%	Reject / escalate for further assessment
Medium	109	29	26.6%	Manual review
Low	114	12	10.5%	Approve

11. Fair-Lending Audit

Using a protected attribute such as sex in a credit decision is prohibited in most jurisdictions, so sex (derived from the personal-status field) is excluded from the model entirely and used only to audit outcomes afterwards. The audit compares approval rates across groups and reports the disparate-impact (DI) ratio — the lower group rate divided by the higher. Approval rates are 54.5% (female) and 62.8% (male), giving a DI ratio of 0.87 (Figure 9), above the common 0.80 "four-fifths" threshold used to flag adverse impact. The model is thus acceptable on fairness as it stands: it neither uses a protected attribute nor produces a strongly uneven outcome. In production the gap would still be monitored over time and investigated if it widened.

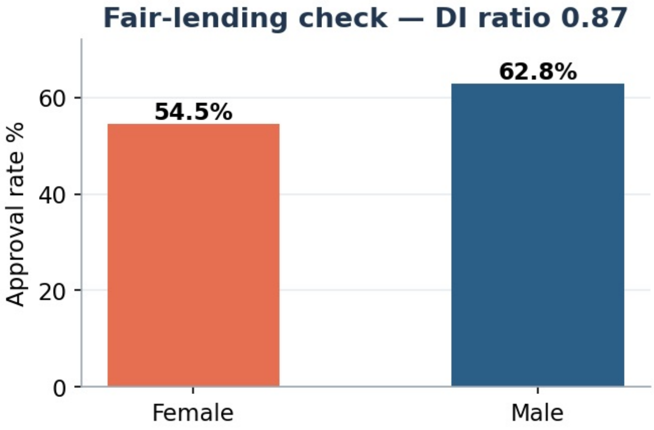

Figure 9 — Approval rate by sex (sex is not a model input); DI ratio 0.87 clears the 0.80 four-fifths line.

12. Limitations and Future Scope

The dataset is small (1,000 rows) and old, so the exact rates here would not carry over to a modern loan book, and results from a 300-row test set come with more uncertainty. There is also no time column, so out-of-time testing is not possible with this data. Useful extensions for future work include: optimal monotonic WoE binning, bootstrap confidence intervals on AUC and KS to put error bars on the results, SHAP-based reason codes for rejection letters, and running the same pipeline on a bigger, more recent dataset such as Home Credit.

13. Conclusion

This project delivers a complete, easy-to-explain credit scorecard. New features were created and checked by Information Value; WoE/IV screening was done with controls for leakage and collinearity; a calibrated logistic model with balanced class weights was trained; and the cut-off was chosen from the dataset's own 5:1 cost rule. The model reaches AUC 0.809, Gini 0.619 and KS 0.514 with a stable PSI of 0.043, performs on par with a gradient-boosting model while staying simple to explain, and passes a four-fifths fair-lending check with the protected attribute left out. Turned into a points-based scorecard with ordered, fully populated risk bands, it supports clear approve / review / reject decisions of the kind used in banks and fintech.

14. Data Source and Citation

Hofmann, H. (1994). *Statlog (German Credit Data)* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5NC77. Licensed CC BY 4.0.
Reproducible artefacts: german_clean.csv, scorecard.csv, iv_table.csv, risk_bands.csv, metrics.json and the Python pipeline.